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THE TIRE SHALL BE IN ACCORDANCE WITH THE APPLICABLE REQUIREMENTS

OF SPECIFICATION MIL-T-5041 EXCEPT AS SPECIFIED HEREIN

| SIZE | PLY RATING | STATIC LOAD RATING LBS. | INFL. PRESS. PSI RATED | BURST PRESS. PSI RATED | BEAD WIDTH IN. MAX. | WEIGHT LBS. MAX. | STATIC UNBAL. OZ-IN MAX. | TREAD DESIGN | MOLD SKID DEPTH MIN. | DEFLECTION % |
|---|---|---|---|---|---|---|---|---|---|---|
| 7.50-10 | 1/12TT | 1800 | 80 | 2/620 | 1.50 | 30.00 | 8 | 3/CHANNEL | .35 | As Determined |

1/ TT TUBE TYPE TIRE
2/ NEW TIRE
3/ TWO CONTINUOS CIRCUMFERENTIAL RIBS

TIRE

INFLATED DIMENSIONS *

| INFLATION PRESSURE (PSI) | CHANNEL RIB OUTSIDE DIA. (INCH) MAX | MIN | SECTION WIDTH (INCH) MAX | MIN | SHOULDER WIDTH MAX @21.60"DIA | RIB/CHANNEL DIMENSIONS |
|---|---|---|---|---|---|---|
| 80 | 23.70 | 22.60 | 7.60 | 7.00 | 6.50 | |
| 120 | 23.70 | 22.60 | 7.60 | 7.00 | 6.50 | SEE FIGURE 1 |
| 155 | 23.70 | 22.60 | 7.60 | 7.00 | 6.50 | |

RIM DIMENSIONS

1. Width between Flanges
2. Ledge Diameter
3. Flange Height
4. Minimum Ledge Width
5. Minimum Flange Width
6. Flange Radius
7. Heel Radius
8. Flange Edge Radius

| 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 |
|---|---|---|---|---|---|---|---|
| 5.50 | 10.00 | .812 | 1.50 | .656 | .469 | .219 | .062 |

* INFLATED PROFILES AT 80 PSI, 120 PSI AND 155 PSI SHALL BE INCLUDED IN THE QUALIFICATION TEST REPORT.

THE TIRE COVERED BY THIS DRAWING SHALL BE SUITABLE FOR USE AND PROVIDE REASONABLE SERVICE ON MILITARY AIRCRAFT DURING ALL NORMAL OPERATIONS AT TAKE OFF SPEEDS OF NOT MORE THAN 100 KNOTS AND LANDING SPEEDS OF NOT MORE THAN 90 KNOTS.

DYNAMIC TEST - THE TIRE SHALL WITHSTAND WITHOUT FAILURE 40 CYCLES OF TEST A, 40 CYCLES OF TEST B, 30 CYCLES OF TEST C, 30 CYCLES OF TEST D, 30 CYCLES OF TEST E AND 30 CYCLES OF TEST F. THE DYNAMIC TEST SEQUENCE SHALL BE A, B, C, D, E, F. THE DYNAMIC TEST INFLATION PRESSURES SHALL BE BASED ON THE FOLLOWING FLAT PLATE INFLATION PRESSURES:

| TEST SPECTRUM | A | B | C | D | E | F |
|---|---|---|---|---|---|---|
| FLAT PLATE INFL (PSI) | 80 | 80 | 120 | 120 | 155 | 155 |

TEST A - TAXI-TAKE-OFF: THE TIRE SHALL BE TAXIED ON THE FLYWHEEL FOR 10,000 FEET UNDER 1,800 POUNDS LOAD AT 30 KNOTS. UPON COMPLETION OF THE TAXI ROLL, THE FLYWHEEL SHALL BE ACCELERATED (SIMULATING TAKE-OFF) AT AN AVERAGE RATE OF 5.1 FEET/SEC/SEC FROM 30 KNOTS TO A SPEED OF 100 KNOTS. THE TIRE SHALL BE UNLANDED AFTER A TAKE-OFF ROLL DISTANCE OF 2525 FEET HAS BEEN COVERED IN 22-24 SECONDS. THE LOAD SHALL BE LINEARLY DECREASED WITH TIME FROM 1,800 POUNDS AT THE BEGINNING OF THE TAKE-OFF ROLE TO 1,000 POUNDS AT THE TIME THE TIRE IS UNLANDED.

TEST B - LANDING TAXI: THE TIRE SHALL BE LANDED AGAINST A FLYWHEEL ROTATING AT A PERIPHERAL SPEED OF 90 KNOTS. IMMEDIATELY UPON LANDING, THE FLYWHEEL SHALL BE DECELERATED AT AN AVERAGE RATE OF 5.8 FEET/SEC/SEC FROM 90 KNOTS TO 30 KNOTS IN 1775 FEET. THE TIRE SHALL THEN BE TAXIED ON THE FLYWHEEL FOR 10,000 FEET UNDER 1800 POUNDS LOAD AT 30 KNOTS. THE TIRE LANDING LOAD SHALL BE 10,500 POUNDS DURING THE FIRST 3 SECONDS OF THE LANDING ROLL AND THEN DECREASED LINEARLY WITH TIME TO 1800 POUNDS DURING THE NEXT 1-2 SECONDS AND REMAIN AT 1800 POUNDS FOR THE DURATION OF THE TEST.

TESTS C AND E - TAXI TAKE OFF: SAME AS TEST A EXCEPT THAT THE TIRE INFLATION PRESSURE SHALL BE ADJUSTED AS NOTED ABOVE.

TESTS D AND F - LANDING TAXI: SAME AS TEST B EXCEPT THAT THE TIRE INFLATION PRESSURE SHALL BE ADJUSTED AS NOTED ABOVE.

WEAR INDICATOR: THE TIRE SHALL INCORPORATE A RED CORD WEAR INDICATOR PLY LOCATED ABOVE THE LAST CARCASS PLY TO ESTABLISH A WEAR LIMIT INDICATOR IN THE CHANNEL CROWN.

FOOTPRINTS: IN ADDITION TO THE FOOTPRINTS REQUIRED BY MIL-T-5041, THE FOLLOWING FOOTPRINTS SHALL BE SUBMITTED WITH THE QUALIFICATION TEST REPORT: 1800 POUNDS @ 80 PSI, 1800 POUNDS @ 120 PSI, 1800 POUNDS @ 155 PSI.

NOTE: THE TIRE CHANNEL AREA SHALL MAKE NO CONTACT WITH THE PLATEN SURFACE AT THE ABOVE FOOTPRING LOAD/INFLATION CONDITIONS.

SECTION: A REPRESENTATIVE SECTION OF THE QUALIFIED TIRE SHALL BE SUBMITTED WITH THE QUALIFICATION TEST REPORT.

QUALIFICATION TEST REPORT: THE QUALIFICATION TEST REPORT SHALL LIST THE RESULTS OF ALL QUALIFICATION TESTS AND THE

APPROVED 3-24-70 REVISED

| P.A. NAVY - AS<br>Other Cust | TITLE<br>TIRE - AIRCRAFT, 7.50-10 CHANNEL TREAD/APPLICATION: OV-10 NOSE LANDING GEAR ONLY | MILITARY STANDARD<br>MS 3502 (AS) |
|---|---|---|
| PROCUREMENT SPECIFICATION<br>MIL-T-5041 | SUPERSEDES | SHEET 1 OF 2 |

DD FORM 1 SEP 67 672-1 N LIMITED COORDINATION PREVIOUS EDITIONS OF THIS FORM ARE OBSOLETE.

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This standard has been approved by the Department of the NAVY and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

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OF SPECIFICATION MIL-T-5041 EXCEPT AS SPECIFIED HEREIN

CONSTRUCTION DETAILS OF THE QUALIFICATION TEST SAMPLE IN THE GENERAL FORM SHOWN IN FIGURE 6 OF MIL-T-5041 WITH DIMENSIONS LISTED AT RATED INFLATION. IN ADDITION IT SHALL ALSO LIST THE ACTUAL TIRE DIMENSIONS WHEN THE TIRE IS INFLATED TO 120 PSI AND 155 PSI. THE REPORT SHALL LIST THE MANUFACTURER'S TEST NUMBER. TWO (2) COPIES OF THE QUALIFICATION TEST REPORT, TOGETHER WITH THE DATA AND MATERIAL SPECIFIED HEREIN, SHALL BE SUBMITTED TO COMMANDER, NAVAL AIR SYSTEMS COMMAND HEADQUARTERS, DEPARTMENT OF THE NAVY, WASHINGTON, D.C. 20360, ATTENTION: AIR-530321.

NOTES: THIS DOCUMENT HAS BEEN PROMULGATED BY THE DEPARTMENT OF DEFENSE AS THE MILITARY STANDARD TO LIMIT THE SELECTION OF THE ITEM, PRODUCT, OR DESIGN COVERED HEREIN IN ENGINEERING, DESIGN AND PROCUREMENT. THIS STANDARD SHALL BECOME EFFECTIVE NOT LATER THAN 90 DAYS AFTER THE LATEST DATE OF APPROVAL SHOWN.

REFERENCE DOCUMENTS SHALL BE OF THE ISSUE IN EFFECT ON DATE OF INVITATIONS FOR BID.

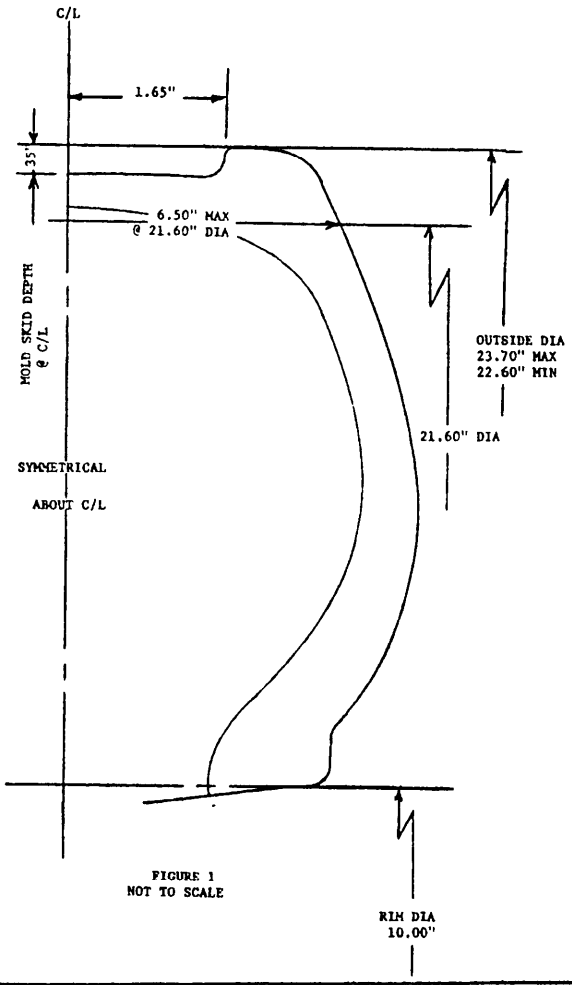



FIGURE 1
NOT TO SCALE

APPROVED 3-24-70 REVISED

This standard has been approved by the Department of the NAVY and is mandatory for use by that activity. All other military activities are required to employ this standard where suitable.

| | | |
|---|---|---|
| P.A. NAVY - AS<br>Other Cust | TITLE<br>TIRE - AIRCRAFT, 7.50-10 CHANNEL TREAD/APPLICATION. OV-10 NOSE LANDING GEAR ONLY | MILITARY STANDARD<br>MS 3502 (AS) |
| PROCUREMENT SPECIFICATION<br>MIL-T-5041 | SUPERSEDES | SHEET 2 OF 2 |

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INCH–POUND

NOTICE
OF VALIDATION

MS3502(AS)
NOTICE 1
29 October 1991

MILITARY STANDARD

# TIRE – AIRCRAFT, 7.50–10 CHANNEL TREAD/APPLICATION: OV–10 NOSE LANDING GEAR ONLY

MS3502(AS), dated 24 March 3970, has been reviewed and determined to be valid for use in acquisition.

Preparing activity:
Navy – AS

AMSC N/A

FSC 2620

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